```
library IEEE;

use IEEE.STD_LOGIC_1164.ALL;
use IEEE.STD_LOGIC_UNSIGNED.ALL;



entity alu4bit is

  port (

      n1 : in STD_LOGIC_VECTOR(3 downto 0);

      n2 : in STD_LOGIC_VECTOR(3 downto 0);

      op : in STD_LOGIC_VECTOR(2 downto 0);

      res : out STD_LOGIC_VECTOR(3 downto 0);
      flag : out STD_LOGIC

  );

end alu4bit;



architecture Behavioral of alu4bit is
```

```
begin

 process(n1, n2, op)

    variable temp : STD_LOGIC_VECTOR(4 downto 0);

 begin



    flag <= '0';



    case op is



     -- Addition

     when "000" =>
 temp := ('0' & n1) + n2;

 res <= temp(3 downto 0);

 flag <= temp(4);
```

```
-- Subtraction

when "001" =>

  if (n1 > n2) then

     res <= n1 - n2;

  else

     res <= n2 - n1;

  end if;



-- AND

when "010" =>

  res <= n1 and n2;



-- OR

when "011" =>
```

```
  res <= n1 or n2;



-- XOR

when "100" =>

  res <= n1 xor n2;




-- NOR

when "101" =>
  res <= n1 nor n2;



-- NOT

when "110" =>
        res <= not n1;



      -- Default

      when others =>
        res <= n2;
```

```
    end case;

  end process;

end Behavioral;
```